GCC Database Solution

User Acceptance Test Plan

Document Date: October 25, 2025

Table of Contents

UAT Objectives

Section 1: Public User Scenarios 5 Tests

Section 2: Analyst User Scenarios 6 Tests

Section 3: Admin User Scenarios 6 Tests

Section 4: External Partner User Scenarios 3 Tests

Section 5: API Integration Scenarios 3 Tests

Section 6: Performance & Load Testing 2 Tests

Section 7: Security & Error Handling 3 Tests

Section 8: Mobile Responsiveness 1 Tests

UAT Sign-Off

Overall Progress

plete
29

UAT Objectives

1. Validate Functionality

Ensure all features work as designed and meet business requirements

2. Ensure Usability

Verify the system is user-friendly and intuitive for all user roles

3. Verify Data Accuracy

Confirm data accuracy, security, and performance standards are met

4. Identify Defects

Discover any defects or issues before production deployment

5. Obtain Sign-Off

Secure stakeholder approval for production release

Test Priority Levels

P0 - CRITICAL

Core functionality that must work for production release

P1 - HIGH

Important features with significant business impact

P2 - MEDIUM

Secondary features that enhance user experience

P3 - LOW

Nice-to-have features or edge cases

Section 1: Public User Scenarios

5 test scenarios

Test Scenario 1.1: View Public Certificate Tiles

Public (Unauthenticated) Certificate Tile View (Landing Page)

P0 - CRITICAL

Scenario Description

A public user visits the GCC website to view available certificates.

Pre-conditions

- Application is running at http://localhost:5101 (or staging URL)
- Database contains sample certificates

Test Steps

1. Open web browser (Chrome, Firefox, or Edge)
2. Navigate to: http://localhost:5101/Certificate/CertificateTileView
3. Observe the certificate tiles display on the page
4. Scroll down to see more certificates (infinite scroll)
5. Verify certificate information is visible: - Certificate number (e.g., "24-0366") - Company name (e.g., "ASSA ABLOY") - Product name (e.g., "Door Hardware") - Certification type (e.g., "FSC Certified") - Status badge (Active, Expired, etc.) - Expiration date

Expected Results

- ☐ Page loads successfully within 2 seconds
- ☐ Certificate tiles display in a grid layout
- ☐ All certificate information is clearly visible and readable
- ☐ Infinite scroll loads more certificates when scrolling down

- [ ] No error messages or broken images appear
- [ ] Mobile responsive (tiles stack on smaller screens)

Actual Results:

Describe what actually happened...

Test Status:

☐ Not Tested ☐ Pass ☐ Fail

Notes/Comments:

Any additional observations or issues...

Test Scenario 1.2: Search for Certificates

Public (Unauthenticated) Autocomplete Search P0 - CRITICAL

Scenario Description

A user searches for a specific certificate, company, or product.

Pre-conditions

- User is on Certificate Tile View page

Test Steps

1. Locate the search box at the top of the page
2. Type "ASSA" into the search box
3. Observe autocomplete dropdown appears with suggestions
4. Verify suggestions are categorized: - Companies (e.g., "ASSA ABLOY") - Products (e.g., "Door Hardware") - Certificate Numbers (e.g., "24-0366")
5. Select "ASSA ABLOY" from the dropdown
6. Verify page filters to show only ASSA ABLOY certificates
7. Clear the search to show all certificates again

Expected Results

- ☐ Search box is clearly visible and accessible
- ☐ Typing displays autocomplete suggestions within 1 second
- ☐ Suggestions are organized by category (Companies, Products, Certificates)
- ☐ Selecting a suggestion filters the certificate list

- ☐ Filtered results are accurate (only matching certificates shown)
- ☐ Search can be cleared to return to full list

Actual Results:

Describe what actually happened...

Test Status:

☐ **Not Tested** ☐ **Pass** ☐ **Fail**

Notes/Comments:

Any additional observations or issues...

Test Scenario 1.3: Filter Certificates by Category

Public (Unauthenticated) Category Filtering P1 - HIGH

Scenario Description

A user filters certificates by certification type, status, or company.

Pre-conditions

- User is on Certificate Tile View page

Test Steps

1. Locate the filter panel on the left side of the page
2. Expand "Certification Type" filter
3. Select "FSC Certified" checkbox
4. Verify page updates to show only FSC certificates
5. Expand "Status" filter
6. Select "Active" checkbox
7. Verify page shows only active FSC certificates
8. Click "Clear Filters" button
9. Verify all certificates are shown again

Expected Results

- ☐ Filter panel is visible and easy to use
- ☐ Selecting filters updates certificate list immediately
- ☐ Multiple filters can be applied simultaneously
- ☐ Filter counts update dynamically (e.g., "FSC Certified (45)")

- [ ] Clear Filters button resets all selections
- [ ] URL updates with filter parameters (shareable link)

Actual Results:

Describe what actually happened...

Test Status:

☐ Not Tested ☐ Pass ☐ Fail

Notes/Comments:

Any additional observations or issues...

Test Scenario 1.4: View Certificate Details

Public (Unauthenticated) Certificate Details Page P0 - CRITICAL

Scenario Description

A user clicks on a certificate to view detailed information.

Pre-conditions

- User is on Certificate Tile View page

Test Steps

1. Click on any certificate tile
2. Verify certificate details page opens
3. Review displayed information: - Certificate number - Company name and logo - Product name and description - Certification type - Effective date - Expiration date - Status - Product IDs (ASIN, UPC, GTIN, etc.)
4. Click "Download PDF" button
5. Verify PDF file downloads to your computer
6. Open the PDF and verify it contains certificate information
7. Click "Back to Search" or browser back button
8. Verify you return to the certificate tile view

Expected Results

- ☐ Certificate details page loads quickly (< 2 seconds)
- ☐ All certificate information is displayed accurately
- ☐ Product IDs are clearly listed (if available)

- [ ] Download PDF button generates a valid PDF file
- [ ] PDF contains professional formatting and all certificate data
- [ ] Navigation back to search works properly

Actual Results:

Describe what actually happened...

Test Status:

☐ Not Tested ☐ Pass ☐ Fail

Notes/Comments:

Any additional observations or issues...

Test Scenario 1.5: Shareable Search URL

Public (Unauthenticated) Shareable URLs P1 - HIGH

Scenario Description

A user creates a filtered search and shares the URL with a colleague.

Pre-conditions

- User is on Certificate Tile View page

Test Steps

1. Apply multiple filters: - Company: "ASSA ABLOY" - Status: "Active" - Certification Type: "FSC Certified"
2. Copy the URL from the browser address bar
3. Open a new browser tab (or incognito window)
4. Paste the copied URL and press Enter
5. Verify the new tab loads with the same filters applied
6. Verify certificate results match the original filtered view

Expected Results

- ☐ URL updates as filters are applied
- ☐ URL contains filter parameters in query string
- ☐ Pasting URL in new tab applies the same filters
- ☐ Filtered results are identical in both tabs
- ☐ URL can be shared via email, chat, etc.

Actual Results:

Describe what actually happened...

Test Status:

☐ Not Tested ☐ Pass ☐ Fail

Notes/Comments:

Any additional observations or issues...

Section 2: Analyst User Scenarios

6 test scenarios

Test Scenario 2.1: Analyst Login

Analyst Authentication P0 - CRITICAL

Scenario Description

An analyst user logs into the system to manage certificates.

Pre-conditions

- Analyst account exists in the system (see UAT User Credentials document)

Test Steps

1. Navigate to http://localhost:5101/Account/Login
2. Enter analyst credentials: - Email: analyst@greencirclecertified.com - Password: Analyst123!
3. Click "Login" button
4. Verify redirect to Dashboard or Certificate Tile View (Internal)
5. Verify user name is displayed in top navigation bar
6. Verify "Logout" option is available

Expected Results

- ☐ Login page loads successfully
- ☐ Valid credentials are accepted
- ☐ User is redirected to internal view after login
- ☐ User name is displayed in navigation
- ☐ Session is maintained (page refresh doesn't log out)

Actual Results:

Describe what actually happened...

Test Status:

☐ Not Tested ☐ Pass ☐ Fail

Notes/Comments:

Any additional observations or issues...

Test Scenario 2.2: Create New Certificate

Analyst Certificate Creation P0 - CRITICAL

Scenario Description

An analyst creates a new product certificate.

Pre-conditions

- Analyst user is logged in
- User is on Certificate Tile View (Internal)

Test Steps

1. Click "Add Certificate" button
2. Verify Create Certificate form opens
3. Fill in required fields: - Certificate Number: "UAT-2025-001" - Company: Select "ASSA ABLOY" from dropdown - Product: Select existing product or create new - Certification Type: Select "FSC Certified" - Effective Date: Select today's date - Expiration Date: Select date 1 year from today - Status: Select "Active"
4. Click "Save" button
5. Verify success message appears
6. Verify redirected to certificate details page or tile view
7. Search for "UAT-2025-001" to confirm it was created

Expected Results

- ☐ Add Certificate button is visible and clickable
- ☐ Create form loads with all required fields
- ☐ Company and Product dropdowns are populated

- ☐ Date pickers work correctly
- ☐ Form validates required fields (error if empty)
- ☐ Certificate is created successfully
- ☐ Success message confirms creation
- ☐ New certificate appears in search results

Actual Results:

Describe what actually happened...

Test Status:

☐ **Not Tested** ☐ **Pass** ☐ **Fail**

Notes/Comments:

Any additional observations or issues...

Test Scenario 2.3: Edit Existing Certificate

Analyst Certificate Editing P0 - CRITICAL

Scenario Description

An analyst updates information on an existing certificate.

Pre-conditions

- Analyst user is logged in
- Certificate "UAT-2025-001" exists (created in previous test)

Test Steps

1. Search for certificate "UAT-2025-001"
2. Click on the certificate tile
3. Click "Edit" button on certificate details page
4. Verify Edit Certificate form loads with existing data pre-filled
5. Update the following: - Product: Change to a different product - Expiration Date: Change to a different date
6. Click "Save Changes" button
7. Verify success message appears
8. Verify updated information is displayed on details page

Expected Results

- ☐ Edit button is visible (Analyst has permission)
- ☐ Edit form loads with current certificate data
- ☐ All fields are editable
- ☐ Changes are saved successfully

- [ ] Success message confirms update
- [ ] Updated data is displayed correctly

Actual Results:

Describe what actually happened...

Test Status:

☐ Not Tested ☐ Pass ☐ Fail

Notes/Comments:

Any additional observations or issues...

Test Scenario 2.4: Product ID Management - Add Product IDs

Analyst Product ID Manager P1 - HIGH

Scenario Description

An analyst adds multiple product IDs (ASINs, UPCs) to a certificate.

Pre-conditions

- Analyst user is logged in
- Certificate "UAT-2025-001" exists

Test Steps

1. Navigate to certificate "UAT-2025-001" details page
2. Click "Edit" button
3. Locate "Product ID Manager" section on the edit form
4. Click "Add Product ID" button
5. Enter the following product IDs one at a time: - ASIN: B08ABC1234 - UPC: 123456789012 - GTIN: 01234567890123
6. Verify each product ID is added to the list
7. Click "Save Changes" to save the certificate
8. Verify product IDs are displayed on certificate details page

Expected Results

- [ ] Product ID Manager section is visible on edit form
- [ ] Add Product ID button works correctly
- [ ] Product ID type can be selected (ASIN, UPC, GTIN, etc.)

- [ ] Product ID value can be entered
- [ ] Multiple product IDs can be added
- [ ] Product IDs are saved with the certificate
- [ ] Product IDs display on certificate details page

Actual Results:

Describe what actually happened...

Test Status:

☐ **Not Tested** ☐ **Pass** ☐ **Fail**

Notes/Comments:

Any additional observations or issues...

Test Scenario 2.5: Product ID Manager - Bulk Import CSV

Analyst Product ID Bulk Import P1 - HIGH

Scenario Description

An analyst imports 100+ ASINs from a CSV file.

Pre-conditions

- Analyst user is logged in
- Certificate "UAT-2025-001" exists
- CSV file with ASINs is prepared (or use template)

Test Steps

1. Navigate to certificate "UAT-2025-001" edit page
2. In Product ID Manager section, click "Import from CSV" button
3. Download the CSV template (if needed)
4. Upload CSV file with the following format: ``` ProductIdType,ProductIdValue ASIN,B08XYZ1111 ASIN,B08XYZ2222 ASIN,B08XYZ3333 ```
5. Click "Upload" or "Import" button
6. Verify import progress indicator appears
7. Verify success message shows number of imported IDs
8. Verify imported product IDs appear in the list
9. Click "Save Changes"
10. Verify all imported product IDs are saved

Expected Results

- ☐ Import from CSV button is visible
- ☐ CSV template can be downloaded
- ☐ CSV file upload works correctly
- ☐ Import validates CSV format
- ☐ Progress indicator shows during import
- ☐ Success message displays import count
- ☐ All valid product IDs are imported
- ☐ Invalid entries show error messages
- ☐ Imported IDs are saved with certificate

Actual Results:

Describe what actually happened...

Test Status:

☐ **Not Tested** ☐ **Pass** ☐ **Fail**

Notes/Comments:

Any additional observations or issues...

Test Scenario 2.6: Export Certificates to CSV

Analyst Certificate Export P1 - HIGH

Scenario Description

An analyst exports filtered certificate data to CSV format.

Pre-conditions

- Analyst user is logged in
- Certificates exist in the database

Test Steps

1. Navigate to Certificate Tile View (Internal)
2. Apply filters: - Company: "ASSA ABLOY" - Status: "Active"
3. Click "Export" button
4. Select "Export to CSV" option
5. Verify CSV file downloads to your computer
6. Open the CSV file in Excel or text editor
7. Verify the following: - CSV contains only filtered certificates (ASSA ABLOY, Active) - All columns are present (Certificate Number, Company, Product, etc.) - Data is formatted correctly (no errors, readable) - Special characters are handled properly

Expected Results

- ☐ Export button is visible
- ☐ Export to CSV option is available
- ☐ CSV file downloads successfully

- ☐ CSV contains filtered data (not all certificates)
- ☐ All certificate fields are included in CSV
- ☐ Data is accurate and properly formatted
- ☐ File can be opened in Excel

Actual Results:

Describe what actually happened...

Test Status:

☐ **Not Tested** ☐ **Pass** ☐ **Fail**

Notes/Comments:

Any additional observations or issues...

Section 3: Admin User Scenarios

6 test scenarios

Test Scenario 3.1: Admin Login and Dashboard Access

Admin Admin Authentication & Dashboard P0 - CRITICAL

Scenario Description

An admin user logs in and accesses the admin dashboard.

Pre-conditions

- Admin account exists in the system

Test Steps

1. Navigate to http://localhost:5101/Account/Login
2. Enter admin credentials: - Email: admin@greencirclecertified.com - Password: Admin123!
3. Click "Login" button
4. Verify redirect to Dashboard
5. Verify Dashboard displays: - Total certificates count - Active certificates count - Expiring certificates count - Recent activity
6. Verify "Admin" menu is visible in navigation
7. Click "Admin" menu and verify options: - User Management - Role Management - Tag Management - Product Management - Company Management - Certificate Types - Partners

Expected Results

- ☐ Admin login successful
- ☐ Dashboard loads with statistics
- ☐ Statistics display accurate data

- ☐ Admin menu is visible (not visible to Analyst)
- ☐ All admin menu options are accessible

Actual Results:

Describe what actually happened...

Test Status:

☐ **Not Tested** ☐ **Pass** ☐ **Fail**

Notes/Comments:

Any additional observations or issues...

Test Scenario 3.2: User Management - Create New User

Admin User Management P0 - CRITICAL

Scenario Description

An admin creates a new user account.

Pre-conditions

- Admin user is logged in

Test Steps

1. Navigate to Admin > User Management
2. Verify user list displays existing users
3. Click "Add New User" button
4. Fill in user details: - Name: "UAT Test User" - Email: "uattest@greencirclecertified.com" - Company: Select "ASSA ABLOY" - Role: Select "Analyst" - Password: "TestUser123!" - Require Password Reset: Check this option
5. Click "Save" button
6. Verify success message appears
7. Verify new user appears in user list
8. Verify user can log in with provided credentials

Expected Results

- ☐ Add New User button is visible
- ☐ User creation form loads correctly
- ☐ All fields are editable

- [ ] Role dropdown is populated
- [ ] Company dropdown is populated
- [ ] User is created successfully
- [ ] User appears in user list
- [ ] New user can log in

Actual Results:

Describe what actually happened...

Test Status:

☐ **Not Tested** ☐ **Pass** ☐ **Fail**

Notes/Comments:

Any additional observations or issues...

Test Scenario 3.3: User Management - Edit User Role

Admin User Role Management P1 - HIGH

Scenario Description

An admin changes a user's role from Analyst to External.

Pre-conditions

- Admin user is logged in
- User "UAT Test User" exists (created in previous test)

Test Steps

1. Navigate to Admin > User Management
2. Find "UAT Test User" in the user list
3. Click "Edit" button for this user
4. Change role from "Analyst" to "External"
5. Click "Save Changes" button
6. Verify success message appears
7. Verify role is updated in user list
8. Log out and log in as "UAT Test User"
9. Verify user now has External role permissions (read-only access)

Expected Results

- □ Edit button works for user
- □ Edit form loads with current user data
- □ Role can be changed via dropdown

  Changes are saved successfully

- [ ] Updated role displays in user list
- [ ] User permissions reflect new role

Actual Results:

Describe what actually happened...

Test Status:

☐ Not Tested ☐ Pass ☐ Fail

Notes/Comments:

Any additional observations or issues...

Test Scenario 3.4: User Management - Delete User

Admin User Deletion P1 - HIGH

Scenario Description

An admin deletes a user account.

Pre-conditions

- Admin user is logged in
- User "UAT Test User" exists

Test Steps

1. Navigate to Admin > User Management
2. Find "UAT Test User" in the user list
3. Click "Delete" button (or Actions dropdown > Delete)
4. Verify confirmation modal appears with warning message
5. Click "Delete" button in the modal
6. Verify success message appears
7. Verify user is removed from the user list
8. Try to log in as the deleted user
9. Verify login fails with appropriate error message

Expected Results

- ☐ Delete button is visible
- ☐ Confirmation modal appears before deletion
- ☐ Modal warns about irreversible action

  User is deleted successfully after confirmation

- ☐ User is removed from user list
- ☐ Deleted user cannot log in
- ☐ Appropriate error message for deleted user login

Actual Results:

Describe what actually happened...

Test Status:

☐ **Not Tested** ☐ **Pass** ☐ **Fail**

Notes/Comments:

Any additional observations or issues...

Test Scenario 3.5: Delete Certificate

Admin Certificate Deletion Admin Only

Scenario Description

An admin deletes a certificate (soft delete).

Pre-conditions

- Admin user is logged in
- Certificate "UAT-2025-001" exists

Test Steps

1. Search for certificate "UAT-2025-001"
2. Open certificate details page
3. Click "Edit" button
4. Verify "Delete Certificate" button is visible (Admin only)
5. Click "Delete Certificate" button
6. Verify confirmation modal appears
7. Click "Delete" in the modal
8. Verify success message appears
9. Verify redirected to certificate tile view
10. Search for "UAT-2025-001"
11. Verify certificate no longer appears in search results
12. (Optional) Verify certificate is soft-deleted in database (IsDeleted = true)

Expected Results

- [ ] Delete button visible only to Admin (not Analyst)
- [ ] Confirmation modal appears before deletion
- [ ] Certificate is deleted successfully
- [ ] Success message confirms deletion
- [ ] Certificate is removed from public view
- [ ] Certificate is soft-deleted (not permanently removed from database)

Actual Results:

Describe what actually happened...

Test Status:

☐ **Not Tested** ☐ **Pass** ☐ **Fail**

Notes/Comments:

Any additional observations or issues...

Test Scenario 3.6: Tag Management

Admin Tag Management P2 - MEDIUM

Scenario Description

An admin creates, edits, and deletes tags.

Pre-conditions

- Admin user is logged in

Test Steps

1. Navigate to Admin > Tag Management
2. Verify existing tags are displayed
3. Click "Add New Tag" button
4. Enter tag details: - Name: "UAT Test Tag" - Tag Type: "Internal" - Description: "Test tag for UAT"
5. Click "Save" button
6. Verify tag is created and appears in list
7. Click "Edit" for the new tag
8. Change Tag Type to "External"
9. Click "Save Changes"
10. Verify tag is updated
11. Click "Delete" for the tag
12. Confirm deletion in modal
13. Verify tag is removed from list

Expected Results

- [ ] Tag Management page loads successfully
- [ ] Existing tags are displayed
- [ ] Add New Tag button works
- [ ] Tag is created successfully
- [ ] Tag can be edited
- [ ] Tag can be deleted with confirmation
- [ ] All changes are saved correctly

Actual Results:

Describe what actually happened...

Test Status:

☐ **Not Tested** ☐ **Pass** ☐ **Fail**

Notes/Comments:

Any additional observations or issues...

Section 4: External Partner User Scenarios

3 test scenarios

Test Scenario 4.1: External User Login

External (Partner) External Authentication P0 - CRITICAL

Scenario Description

An external partner (e.g., Amazon) logs in to view their certificates.

Pre-conditions

- External user account exists

Test Steps

1. Navigate to http://localhost:5101/Account/Login
2. Enter external user credentials: - Email: partner@amazon.com - Password: Partner123!
3. Click "Login" button
4. Verify redirect to Certificate Tile View (External)
5. Verify only certificates associated with Amazon partner are displayed
6. Verify user name and company are displayed in navigation
7. Verify "Edit" and "Delete" buttons are NOT visible on certificates

Expected Results

- ☐ External user login successful
- ☐ User sees only partner-specific certificates (Amazon filter applied)
- ☐ User has read-only access (no create, edit, delete)
- ☐ User interface is simplified for external users
- ☐ No admin menu options visible

Actual Results:

Describe what actually happened...

Test Status:

☐ Not Tested ☐ Pass ☐ Fail

Notes/Comments:

Any additional observations or issues...

Test Scenario 4.2: External User - View and Download Certificates

External (Partner) Certificate Viewing and Download P0 - CRITICAL

Scenario Description

An external user views and downloads their certificates.

Pre-conditions

- External user is logged in (partner@amazon.com)

Test Steps

1. Verify certificate tile view shows only Amazon-associated certificates
2. Click on a certificate tile
3. Verify certificate details page opens
4. Review certificate information
5. Click "Download PDF" button
6. Verify PDF downloads successfully
7. Open PDF and verify it contains certificate data
8. Verify "Edit" button is NOT visible (read-only access)

Expected Results

- ☐ External user sees only their company's certificates
- ☐ Certificate details page loads correctly
- ☐ All certificate information is visible
- ☐ Download PDF button works

- ☐ PDF is generated successfully
- ☐ No edit or delete options are available

Actual Results:

Describe what actually happened...

Test Status:

☐ **Not Tested** ☐ **Pass** ☐ **Fail**

Notes/Comments:

Any additional observations or issues...

Test Scenario 4.3: External User - Export Certificates

External (Partner) Certificate Export P1 - HIGH

Scenario Description

An external user exports their certificates to CSV.

Pre-conditions

- External user is logged in

Test Steps

1. Navigate to Certificate Tile View (External)
2. Verify only partner-specific certificates are visible
3. Click "Export" button
4. Select "Export to CSV"
5. Verify CSV file downloads
6. Open CSV file
7. Verify CSV contains only partner's certificates
8. Verify all required columns are present

Expected Results

- ☐ Export button is visible to external users
- ☐ Export to CSV option works
- ☐ CSV contains only partner-specific certificates
- ☐ CSV is formatted correctly
- ☐ External user cannot export other companies' data

Actual Results:

Describe what actually happened...

Test Status:

☐ Not Tested ☐ Pass ☐ Fail

Notes/Comments:

Any additional observations or issues...

Section 5: API Integration Scenarios

3 test scenarios

Test Scenario 5.1: API - Authenticate with API Key

API Consumer (Developer/Partner) | API Authentication | P0 - CRITICAL

Scenario Description

An API consumer authenticates using an API key.

Pre-conditions

- API key exists for testing (see UAT User Credentials document)
- API testing tool (Postman, Insomnia, or curl) is available

Test Steps

1. Open API testing tool (e.g., Postman)
2. Create a new GET request to: ``` http://localhost:5101/api/CertificateTileExternal/tiles ```
3. Add header: - Key: X-API-Key - Value: [Use API key from credentials document]
4. Send the request
5. Verify response: - Status: 200 OK - Body: JSON array of certificates
6. Remove the X-API-Key header
7. Send request again
8. Verify response: - Status: 401 Unauthorized

Expected Results

- ☐ API request with valid API key returns 200 OK
- ☐ Response contains JSON array of certificates
- ☐ API request without API key returns 401 Unauthorized

- [ ] Error message is clear and helpful

Actual Results:

Describe what actually happened...

Test Status:

☐ Not Tested ☐ Pass ☐ Fail

Notes/Comments:

Any additional observations or issues...

Test Scenario 5.2: API - Get Partner-Specific Certificates

API Consumer (Amazon Partner) Partner Filtering P0 - CRITICAL

Scenario Description

Amazon partner API returns only Amazon-associated certificates.

Pre-conditions

- Amazon partner API key is configured

Test Steps

1. Open API testing tool
2. Create GET request to: ``` http://localhost:5101/api/CertificateTileExternal/tiles ```
3. Add Amazon partner API key in X-API-Key header
4. Send request
5. Review response JSON
6. Verify all returned certificates have Partner field = "Amazon" (or are associated with Amazon)
7. Verify certificates from other partners (Wayfair, etc.) are NOT included

Expected Results

- ☐ API returns only Amazon-associated certificates
- ☐ Response does not include other partners' certificates
- ☐ JSON structure is correct
- ☐ All required certificate fields are present

- [ ] Response time is acceptable (< 2 seconds)

Actual Results:

Describe what actually happened...

Test Status:

☐ Not Tested ☐ Pass ☐ Fail

Notes/Comments:

Any additional observations or issues...

Test Scenario 5.3: API - Rate Limiting

API Consumer API Rate Limiting P2 - MEDIUM

Scenario Description

API enforces rate limiting to prevent abuse.

Pre-conditions

- API key is configured
- Rate limit is set (e.g., 1000 requests per hour)

Test Steps

1. Use API testing tool or script
2. Make 1000 API requests rapidly (within a few minutes)
3. Verify first 1000 requests succeed (200 OK)
4. Make the 1001st request
5. Verify response: - Status: 429 Too Many Requests - Body: Error message explaining rate limit exceeded
6. Wait for rate limit window to reset (1 hour)
7. Make another request
8. Verify request succeeds (200 OK)

Expected Results

- ☐ API allows up to 1000 requests per hour
- ☐ Request 1001 returns 429 Too Many Requests
- ☐ Error message explains rate limit
- ☐ Rate limit resets after time window

- [ ] Requests succeed after reset

Actual Results:

Describe what actually happened...

Test Status:

☐ Not Tested ☐ Pass ☐ Fail

Notes/Comments:

Any additional observations or issues...

Section 6: Performance & Load Testing

2 test scenarios

Test Scenario 6.1: Page Load Performance

Any Application Performance P1 - HIGH

Scenario Description

Verify key pages load within acceptable time limits.

Pre-conditions

- Application is running
- Database contains 180+ certificates

Test Steps

1. Clear browser cache
2. Navigate to Certificate Tile View
3. Measure page load time (use browser Developer Tools > Network tab)
4. Record load time
5. Repeat for other pages: - Certificate Details - Dashboard - Admin > User Management
6. Verify all pages load within target times: - Certificate Tile View: < 2 seconds - Certificate Details: < 1.5 seconds - Dashboard: < 2 seconds - Admin pages: < 2 seconds

Expected Results

- ☐ Certificate Tile View loads in < 2 seconds
- ☐ Certificate Details loads in < 1.5 seconds
- ☐ Dashboard loads in < 2 seconds
- ☐ Admin pages load in < 2 seconds

No JavaScript errors in browser console

- [ ] Images and resources load properly

Actual Results:

Describe what actually happened...

Test Status:

☐ Not Tested ☐ Pass ☐ Fail

Notes/Comments:

Any additional observations or issues...

Test Scenario 6.2: Infinite Scroll Performance

Public Infinite Scroll P1 - HIGH

Scenario Description

Verify infinite scroll loads additional certificates smoothly.

Pre-conditions

- Certificate Tile View is loaded
- Database contains 180+ certificates

Test Steps

1. Open Certificate Tile View
2. Observe initial load (first 20-30 certificates)
3. Scroll down to bottom of page
4. Verify loading indicator appears
5. Measure time for next batch to load
6. Verify next batch (20-30 more certificates) loads within 1 second
7. Continue scrolling and loading until all certificates are visible
8. Verify no errors or slowdowns

Expected Results

- ☐ Initial batch loads quickly (< 2 seconds)
- ☐ Loading indicator appears when scrolling to bottom
- ☐ Next batch loads within 1 second
- ☐ Scrolling is smooth (no lag or freezing)
- ☐ All 180+ certificates load successfully

- [ ] No duplicate certificates appear

Actual Results:

Describe what actually happened...

Test Status:

☐ Not Tested ☐ Pass ☐ Fail

Notes/Comments:

Any additional observations or issues...

Section 7: Security & Error Handling

3 test scenarios

Test Scenario 7.1: Unauthorized Access Prevention

Unauthenticated User | Authorization & Access Control | P0 - CRITICAL

Scenario Description

Verify unauthenticated users cannot access protected pages.

Pre-conditions

- User is NOT logged in (logout if necessary)

Test Steps

1. Try to access protected pages directly via URL: - http://localhost:5101/Certificate/Create - http://localhost:5101/Certificate/Edit/1 - http://localhost:5101/Admin/UserManagement - http://localhost:5101/Dashboard
2. For each page, verify: - Page redirects to /Account/Login - User is prompted to log in - After login, user is redirected to intended page (if authorized)

Expected Results

- ☐ All protected pages redirect to login page
- ☐ User cannot access pages without authentication
- ☐ After login, user is redirected to originally requested page
- ☐ Unauthorized users (e.g., Analyst accessing Admin pages) see 403 Forbidden

Actual Results:

Describe what actually happened...

Test Status:

☐ Not Tested ☐ Pass ☐ Fail

Notes/Comments:

Any additional observations or issues...

Test Scenario 7.2: Invalid Certificate Number Handling

Any Error Handling P2 - MEDIUM

Scenario Description

User tries to view a certificate that doesn't exist.

Pre-conditions

- User is on Certificate Tile View

Test Steps

1. Manually navigate to a non-existent certificate: ``` http://localhost:5101/Certificate/Details/99-9999 ```
2. Verify error page is displayed (404 Not Found)
3. Verify error message is user-friendly: - "Certificate not found" or similar message - Suggestion to return to search
4. Click "Back to Search" link
5. Verify user is redirected to Certificate Tile View

Expected Results

- ☐ 404 error page is displayed
- ☐ Error message is clear and user-friendly
- ☐ No technical error details exposed to user
- ☐ Navigation link back to search is provided
- ☐ Application does not crash

Actual Results:

Describe what actually happened...

Test Status:

☐ Not Tested ☐ Pass ☐ Fail

Notes/Comments:

Any additional observations or issues...

Test Scenario 7.3: Form Validation

Analyst Input Validation P1 - HIGH

Scenario Description

Verify form validation prevents invalid data entry.

Pre-conditions

- Analyst user is logged in
- User is on Create Certificate form

Test Steps

1. Click "Save" button without filling any fields
2. Verify validation errors appear for required fields: - Certificate Number - Company - Product - Effective Date - Expiration Date
3. Fill in Certificate Number: "TEST-001"
4. Fill in Effective Date: "2025-10-22"
5. Fill in Expiration Date: "2025-01-01" (earlier than effective date)
6. Click "Save" button
7. Verify validation error: "Expiration date must be after effective date"
8. Correct Expiration Date to "2026-10-22"
9. Fill in remaining required fields
10. Click "Save"
11. Verify certificate is created successfully

Expected Results

- ☐ Form validates required fields before submission

- [ ] Validation errors are displayed clearly
- [ ] Date logic is validated (expiration > effective)
- [ ] User cannot submit invalid data
- [ ] Valid data is accepted and saved

Actual Results:

Describe what actually happened...

Test Status:

☐ **Not Tested** ☐ **Pass** ☐ **Fail**

Notes/Comments:

Any additional observations or issues...

Section 8: Mobile Responsiveness

1 test scenarios

Test Scenario 8.1: Mobile View - Certificate Tiles

Public Mobile Responsive Design P2 - MEDIUM

Scenario Description

Verify certificate tile view works on mobile devices.

Pre-conditions

- Application is running
- Mobile device or browser responsive mode is available

Test Steps

1. Open browser Developer Tools (F12)
2. Enable device toolbar (mobile emulation)
3. Select iPhone 12 (or similar mobile device)
4. Navigate to Certificate Tile View
5. Verify layout adjustments: - Tiles stack vertically (single column) - Filter panel collapses into expandable menu - Navigation menu becomes hamburger menu - Text is readable (16px minimum font size) - Buttons are touch-friendly (44px minimum)
6. Try scrolling, filtering, and searching
7. Verify all functionality works on mobile
8. Test on tablet size (iPad)
9. Verify layout adapts appropriately

Expected Results

- ☐ Layout is responsive and adapts to screen size

- ☐ Tiles stack vertically on mobile
- ☐ Filter panel is accessible via hamburger menu
- ☐ All text is readable (no tiny fonts)
- ☐ Buttons are large enough to tap easily
- ☐ Infinite scroll works on mobile
- ☐ Search functionality works on mobile
- ☐ No horizontal scrolling required

Actual Results:

Describe what actually happened...

Test Status:

☐ **Not Tested** ☐ **Pass** ☐ **Fail**

Notes/Comments:

Any additional observations or issues...

Document Created: October 25, 2025